

Andrew Shao

shaojh@umich.edu | <https://www.linkedin.com/in/andrewshao628> | <https://andrewshao628.github.io>

EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Science in Engineering: Computer Science

Aug 2022 - Dec 2025

• Relevant Coursework:

Web Systems, Operating Systems, Distributed Systems, Applied Parallel Programming with GPUs, Computer Organization, Machine Learning, Natural Language Processing, Applied Linear Algebra

WORK EXPERIENCE

Rockwell Automation

Mayfield Heights, OH

Software Intern

May 2025 - Aug 2025

- Designed and deployed an asynchronous event-driven pipeline using Apache Kafka and Python to process real-time telemetry data for debug purposes from virtualized controllers, increasing data ingestion throughput by 35%.
- Navigated a large legacy codebase for Logix Designer to resolve 10+ critical concurrency bugs in C++ and C; utilized .NET profiling through Visual Studio optimizing asynchronous resource management, reducing interface latency.
- Built an automated security tool in Python using GitLab CI/CD that integrates with the BlackDuck API to parse security vulnerability data and generate Jira tickets, streamlining the reporting workflow and improving triage efficiency by 20%.

Sigwise

Solon, OH

Software Intern

May 2023 - Jul 2023

- Developed a video storage Android application using Java, implementing an LRU caching mechanism and integrating with the Google Cloud Storage (GCS) API to ensure low-latency video playback, reducing load times by 20%.
- Designed and deployed Docker containers for backend alert services, utilizing Kubernetes for cluster management, scaling, and creating consistent development environments to streamline testing.

Hyland Software

Westlake, OH

Software Intern (PEEKE Internship Program)

Jun 2021 - Sept 2021

- Created a full-stack web application to host the next Hyland Hackathon using React.js, Django, and SQLite, designed to display real-time announcements, chat messages, and team data, allowing the hackathon to shift to a hybrid format.
- Developed and executed unit and integration test cases using Pytest and Jest to ensure code quality and reliability.

PROJECTS

S-Watch: Online Movie Streaming Platform

Golang, Gin-gonic, ReactJS, Javascript, GCP, MongoDB, OpenAI

- Developed a full-stack movie streaming platform using a Go/Gin backend and React frontend, featuring secure JWT authentication, responsive UI, and an AI recommendation engine built with LangChainGo and OpenAI.
- Integrated MongoDB Atlas for scalable data storage and deployed the application using GCP Cloud Run for the backend and Vercel for the frontend to ensure high availability and seamless media playback.

Distributed Key-Value Store (Paxos & Sharded KV Store)

C++, C

- Built a fault-tolerant, horizontally scalable distributed key-value store in C++ using Paxos consensus and a sharded architecture, ensuring linearizable consistency and live shard migration.
- Designed a resilient RPC communication layer with idempotent request handling and optimized routing to guarantee at-most-once semantics and minimize end-to-end latency.

MapReduce Framework & Search Engine

Python, Flask, ReactJS, Javascript, Java, HTML, CSS

- Developed a distributed MapReduce framework in Python using socket programming, multithreading, and a Manager-Worker architecture to enable fault-tolerant parallel data processing with worker monitoring and dynamic load balancing.
- Built a multi-stage data mining pipeline that processed thousands of crawled HTML documents into an inverted index, computing tf-idf, document normalization, and PageRank metrics for scalable search engine information retrieval.

SKILLS

Languages: C++, Go, C, Python, Java, C, SQL, Bash, JavaScript

Technologies & Cloud: Docker, Kubernetes, AWS, GCP, CUDA, Linux/Unix, CI/CD (Jenkins), Kafka, Flask, React